

multi_v1

Start of Block: Consent

Q1

Informed Consent Form IRB-FY2020-4512 Title: Sources of Difficulty in Language Processing and Learning Principal Investigator: Tal Linzen (Department of Linguistics and the Center for Data Science, New York University) PURPOSE OF RESEARCH STUDY: The goal of the project is to measure what makes particular aspects of language easier or harder to learn and understand. PROCEDURES: You will be asked to read or listen to language, and answer questions about what you've read or heard. The sentences may be in English or in a made-up language that you will learn during the experiment. The experiment involves a single session that will take up to an hour; there will be up to five sessions, but most participants will only participate in a single session. RISKS/DISCOMFORTS: The risks associated with participation in this study are no greater than those encountered in daily life. BENEFITS: There are no direct benefits to you from participating in this study. This study may benefit society if the results lead to a better understanding of what makes certain aspects of language easier or harder to learn and understand. VOLUNTARY PARTICIPATION AND RIGHT TO WITHDRAW: Your participation in this study is entirely voluntary: You choose whether to participate. If you decide not to participate, there are no penalties, and you will not lose any benefits to which you would otherwise be entitled. If you choose to participate in the study, you can stop your participation at any time, without any penalty or loss of benefits. CONFIDENTIALITY: Any study records that identify you will be kept confidential to the extent possible by law. The records from your participation may be reviewed by people responsible for making sure that research is done properly. Otherwise, records that identify you will be available only to people working on the study, unless you give permission for other people to see the records. Any study records that include your name will be kept in a password-protected database. On all records of test results we will use a code number rather than your name. COMPENSATION: You will receive compensation in proportion to the length of the session. IF YOU HAVE QUESTIONS OR CONCERNS: You can ask questions about this research study now or at any time during the study, by talking to the researcher(s) working with you or by emailing Dr. Tal Linzen at linzen@nyu.edu. If you have questions about your rights as a research participant or feel that you have not been treated fairly, please contact the NYU Institutional Review Board at ask.humansubjects@nyu.edu or **212.998.4808**.



Q2 Do you consent to participate in this study?

- ☐ I consent (1)
- ☐ I do not consent (2)

End of Block: Consent

Start of Block: Audio Setup



Q1 Before beginning, let's ensure your audio is working! Make sure that you can hear the following audio clip, then select the appropriate answer(s) below.

- ☐ 412 (4)
- ☐ 28 (5)
- ☐ -356 (6)
- ☐ 7 (7)

End of Block: Audio Setup

Start of Block: Instructions

Q1 Instructions Please complete this task in one sitting **without writing anything down** or using outside resources. You will listen to four passages of text. Each passage will be between 1-4 minutes long. You will only be able to listen to each passage **once**. After listening to each passage, you will answer some questions about its contents. Do your best to answer accurately, but don't worry if you are unsure about any answers. Some passages and questions may be more difficult to answer than others. You may take a short break between passages if needed, but please keep breaks brief and stay focused on the task.

End of Block: Instructions

Start of Block: birds_passage

birds_timing_passage Timing

First Click (1)

Last Click (2)

Page Submit (3)

Click Count (4)



birds_trial Loading...

End of Block: birds_passage

Start of Block: birds_qs - Jun 29, 2026

QB_timing Timing

First Click (1)

Last Click (2)

Page Submit (3)

Click Count (4)



QB1 Which of the following statements accurately describes the ashen skreel's plumage?
Select all that apply.

- ☐ Mostly gray, with a lighter patch at the throat (1)
- ☐ Faint streaking along the back (2)
- ☐ A reddish wash across the breast (3)
- ☐ Rust orange, with a lighter patch at the throat (4)
- ☐ None of the above (5)



QB2 Which of the following does the Copperhook finch eat? Select all that apply.

- ☐ Fruit (1)
 - ☐ Seeds (2)
 - ☐ Grubs (3)
 - ☐ Nectar (4)
 - ☐ None of the above (5)
-



QB3 Which of the following correctly describes a difference between the two birds? Select all that apply.

- ☐ The ashen skreel eats carrion, while the Copperhook finch eats seeds (1)
 - ☐ The ashen skreel is mostly gray, whereas the Copperhook finch is mostly orange (2)
 - ☐ The Copperhook finch eats seeds, unlike the fruit-eating ashen skreel (3)
 - ☐ The ashen skreel has black wingtips, while the Copperhook finch has a pale throat (4)
 - ☐ None of the above (5)
-



QB4 Which of the following correctly matches the bird to its Latin name? Select all that apply.

- ☐ Ashen skreel -- *Fringilla rufa* (1)
 - ☐ Copperhook finch -- *Plumbea rufa* (2)
 - ☐ Copperhook finch -- *Fringilla vorax* (3)
 - ☐ Ashen skreel -- *Plumbea rufa* (4)
 - ☐ None of the above (5)
-



QB5 Which of the following accurately describes the Copperhook finch's plumage? Select all that apply.

- ☐ A rust-orange body (1)
 - ☐ Dark tips to the wings (2)
 - ☐ A bright yellow crest (3)
 - ☐ Bluish bars across the tail (4)
 - ☐ None of the above (5)
-



QB_A Which of the following describes the two birds? This is an attention check question.
Select only the option that mentions two colors.

- ☐ Residency versus migration (1)
- ☐ Solitary defence versus gregarious flocking (2)
- ☐ Predator versus prey (3)
- ☐ Green versus black (4)
- ☐ None of the above (5)

End of Block: birds_qs - Jun 29, 2026

Start of Block: birds_feedback

QB_passage How difficult did you find the passage and questions?

	0	1	2	3	4	5	6	7	8	9	10
Passage (1)											
Questions (2)											

QB_comment Is there anything about the study you found notable or wanted to let us know about? For example: comments about the passage and/or narration, how challenging the questions were, anything you notice about yourself while doing the task, or any distractions or technical issues.

End of Block: birds_feedback

Start of Block: Break

Q53 You have completed a passage. You may take a short break before continuing. When you are ready, click the button below to proceed.

End of Block: Break

Start of Block: fruits_passage

fruits_timing_passage Timing

First Click (1)

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Page Submit (3)

Click Count (4)



fruits_trial Loading...

End of Block: fruits_passage

Start of Block: fruits_qs

QF_timing Timing

First Click (1)

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Page Submit (3)

Click Count (4)



QF1 Which of the following correctly describes the Tawnberry's flesh? Select all that apply.

- ☐ Deep purple (1)
 - ☐ Orange-pink (2)
 - ☐ Translucent near the core (3)
 - ☐ Whitish, similar to a blueberry (4)
 - ☐ None of the above (5)
-



QF2 Which of the following accurately describes the taste of the Blisterfig? Select all that apply.

- ☐ Peppery aftertaste (1)
 - ☐ Sweet (2)
 - ☐ A warm spicy flavor (3)
 - ☐ Crisp and slightly sour (4)
 - ☐ None of the above (5)
-



QF3 Which of the following correctly describes a difference between the two fruits? Select all that apply.

- ☐ The Tawnberry has purple flesh, while the Blisterfig has orange-pink flesh (1)
 - ☐ The Blisterfig has a spicy-sweet taste, while the Tawnberry is rich and jammy (2)
 - ☐ The Blisterfig has purple flesh, while the Tawnberry is orange-pink (3)
 - ☐ The Tawnberry ripens in June to July, while the Blisterfig ripens in August to September (4)
 - ☐ None of the above (5)
-



QF4 In which of the following months are both tawnberries and blisterfigs available? Select all that apply.

- ☐ May (1)
 - ☐ June (2)
 - ☐ July (3)
 - ☐ August (4)
 - ☐ None of the above (5)
-



QF5 Which of the following accurately describes the Tawnberry's flavor? Select all that apply.

- ☐ Rich and sweet (1)
- ☐ A warm spicy note (2)
- ☐ Jammy finish (3)
- ☐ Mild and watery (4)
- ☐ None of the above (5)



QF_A Which of the following describes the winged fig incorrectly? This is an attention check. Select only the option that names a color mentioned in the passage.

- ☐ Grows in clusters (1)
- ☐ Woodland edge habitat (2)
- ☐ Purple (3)
- ☐ Spicy and sweet (4)
- ☐ None of the above (5)

End of Block: fruits_qs

Start of Block: fruits_feedback

QF_passage How difficult did you find the passage and questions?

		Easy	Difficult									
		0	1	2	3	4	5	6	7	8	9	10
Passage (1)												
Questions (2)												

QF_comment Is there anything about the study you found notable or wanted to let us know about? For example: comments about the passage and/or narration, how challenging the questions were, anything you notice about yourself while doing the task, or any distractions or technical issues.

End of Block: fruits_feedback

Start of Block: astronomy_passage

astronomy_timing_passage Timing
First Click (1)
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Page Submit (3)
Click Count (4)



astronomy_trial Loading...

End of Block: astronomy_passage

Start of Block: astronomy_qs - Jun 29, 2026

QA_timing Timing
First Click (1)
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Page Submit (3)
Click Count (4)



QA1 What do astronomers study the Pictor Nebula for? Select all that apply.

- ☐ How new stars take shape inside clouds of gas (1)
 - ☐ How star formation stops happening (2)
 - ☐ How stars of the same age grow old together (3)
 - ☐ How a dense star pulls material from a nearby companion (4)
 - ☐ None of the above (5)
-



QA2 Which of the following correctly describes the location of the Aquila Pulsar relative to Earth? Select all that apply.

- ☐ The Aquila Pulsar is about six thousand light-years away from Earth (1)
 - ☐ The Aquila Pulsar is about four thousand light-years away from Earth (2)
 - ☐ The Aquila Pulsar is closer to the center of the galaxy than Earth (3)
 - ☐ The Aquila Pulsar is further from the center of the galaxy than Earth (4)
 - ☐ None of the above (5)
-



QA3 Which of the following correctly describes differences between the Pictor Nebula and the Aquila Pulsar? Select all that apply.

- ☐ The Pictor Nebula is closer to Earth than the Aquila Pulsar (1)
 - ☐ The Pictor Nebula is studied to understand star formation, while the Aquila Pulsar is observed to understand how gravitational waves affect black holes (2)
 - ☐ The Pictor Nebula is observed with optical telescopes in red light, while the Aquila Pulsar can only be detected by radio telescopes (3)
 - ☐ The Aquila Pulsar is closer to Earth than the Pictor Nebula (4)
 - ☐ None of the above (5)
-



QA4 Which of the following correctly matches each object to its location? Select all that apply.

- ☐ Pictor Nebula — about six thousand light-years away, toward the galactic center (1)
 - ☐ Aquila Pulsar — about four thousand light-years away, between spiral arms (2)
 - ☐ Pictor Nebula — about twenty-five thousand light-years away, near the outer halo (3)
 - ☐ Aquila Pulsar — about three thousand light-years away, in a dense star field (4)
 - ☐ None of the above (5)
-



QA5 Which of the following correctly describes how the Aquila Pulsar is observed? Select all that apply.

- ☐ It is observed using radio telescopes (1)
- ☐ It is observed using optical telescopes sensitive to red-light emission (2)
- ☐ It is observed with the naked eye under dark skies (3)
- ☐ is observed to measure how its size changes over time (4)
- ☐ None of the above (5)



QA_A Which of the following topics are uncovered in the passage? If you are paying attention, select only "what the objects are used to study".

- ☐ Where the objects were discovered (1)
- ☐ What the objects are used to study (2)
- ☐ How old the objects are (3)
- ☐ Who first observed the objects (4)
- ☐ None of the above (5)

End of Block: astronomy_qs - Jun 29, 2026

Start of Block: astronomy_feedback

QA_passage How difficult did you find the passage and questions?

Easy					Difficult					
0	1	2	3	4	5	6	7	8	9	10

Passage (1)	
Questions (2)	

QA_comment Is there anything about the study you found notable or wanted to let us know about? For example: comments about the passage and/or narration, how challenging the questions were, anything you notice about yourself while doing the task, or any distractions or technical issues.

End of Block: astronomy_feedback

Start of Block: musical_instruments_passage

musical_instruments_timing_passage Timing
First Click (1)
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musical_instruments_trial Loading...

End of Block: musical_instruments_passage

Start of Block: musical_instruments_qs - Jun 29, 2026

QI_timing Timing

First Click (1)

Last Click (2)

Page Submit (3)

Click Count (4)



QI1 Which of the following correctly describes the Belvero's body shape? Select all that apply.

- ☐ It has a deep rounded belly (1)
- ☐ It has a flat, rectangular body (2)
- ☐ It has a pear-shaped body (3)
- ☐ It is shaped like a lyre (4)
- ☐ None of the above (5)



QI2 Which of the following correctly describes the sound of the Kasten-zither? Select all that apply.

- ☐ Warm and full, sustaining and carrying across a room (1)
- ☐ Bright and dry, fading quickly after each pluck (2)
- ☐ A rich buzzing tone, like a horn (3)
- ☐ Deep and resonant, with a short decay (4)
- ☐ None of the above (5)



Q13 Which of the following correctly describes both the Belvero and the Kasten-zither? Select all that apply.

- ☐ The Belvero has a flat, shallow body, while the Kasten-zither makes a bright sound that fades quickly (1)
 - ☐ The Belvero produces a pear-shaped body, while the Kasten-zither makes a bright sound that fades quickly (2)
 - ☐ The Kasten-zither has a flat, shallow body, while the Belvero produces a warm, full tone (3)
 - ☐ The Belvero sounds bright and dry, while the Kasten-zither produces a warm, full tone (4)
 - ☐ None of the above (5)
-



Q14 Which of the following correctly describes the origins of the two instruments? Select all that apply.

- ☐ The Belvero was documented before the Kasten-zither (1)
 - ☐ The Kasten-zither was first documented by Gunther Kasten (2)
 - ☐ The Belvero was first documented by Alonso Belver (3)
 - ☐ The Kasten-zither was documented before the Belvero (4)
 - ☐ None of the above (5)
-



QI5 Which of the following accurately describes the sound of the Belvero? Select all that apply.

- ☐ It produces a warm, full tone (1)
- ☐ Its sound carries well across a room (2)
- ☐ Its sound is bright and dry (3)
- ☐ It produces a rich buzzing tone, like a horn (4)
- ☐ None of the above (5)



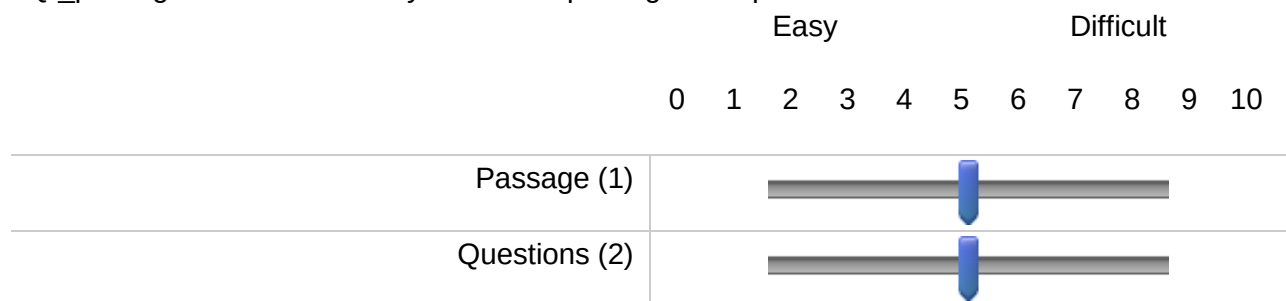
QI_A Which of the following describes a similarity between the two instruments? This is an attention check question. Select only the option that contains a year.

- ☐ Two instruments from opposite ends of the world (1)
- ☐ One instrument played upright, the other laid flat (2)
- ☐ First documented in 1444, the Belvero is the older of the two (3)
- ☐ One instrument is plucked with fingernails, the other with a pick (4)
- ☐ None of the above (5)

End of Block: musical_instruments_qs - Jun 29, 2026

Start of Block: instruments_feedback

QI_passage How difficult did you find the passage and questions?



Q1_comment Is there anything about the previous task you found notable or wanted to let us know about? For example: comments about the passage and/or narration, how challenging the questions were, anything you notice about yourself while doing the task, or any distractions or technical issues.

End of Block: instruments_feedback

Start of Block: Debrief

Q1 Thank you for participating! Your responses have been recorded. Please click the 'End Task' button below to return to Prolific and confirm your completion. **Do not close this window or use the back button** — your submission is only finalized once you are redirected to Prolific.

End of Block: Debrief
